

MCS-1: Probability and Statistics (MA6.302)

Monsoon 2026

Lecture 2

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- Our conditional probability is still describing “the probability of something”, so we expect it to behave like a probability distribution.

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- What do you think $P(A|B)$ should be?

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- A conditional probability is only defined if $P(B) > 0$.

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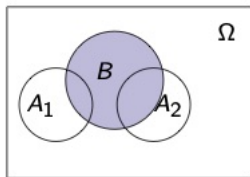
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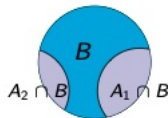
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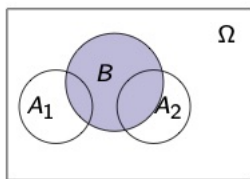


Conditioning on B

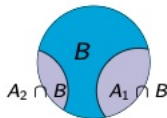


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$$\begin{aligned} P(A_1 \cup A_2|B) &= \frac{P((A_1 \cup A_2) \cap B)}{P(B)} = \frac{P((A_1 \cap B) \cup (A_2 \cap B))}{P(B)} \\ &= \frac{P(A_1 \cap B) + P(A_2 \cap B)}{P(B)} = \frac{P(A_1 \cap B)}{P(B)} + \frac{P(A_2 \cap B)}{P(B)} \\ &= P(A_1|B) + P(A_2|B) \end{aligned}$$

Properties of Conditional Probability

If $P(B) > 0$,

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Consider the experiment of tossing a fair coin three times. What is the probability of getting alternating heads and tails conditioned on the event that your first toss gives a head?

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- **Interpretation of $P(A|B)$:** Your new world/sample space when you condition is $B = \{HHT, HTH, HTT, HHH\}$. Each of these are equally likely. Out of these 1 event satisfies alternating heads and tails. So, $P(A|B) = 1/4$.

Total Probability Theorem

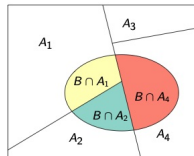
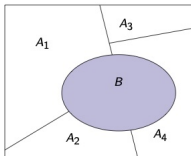
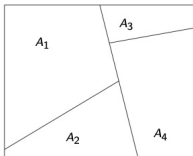
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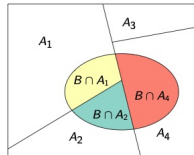
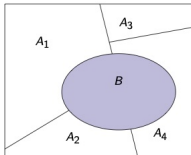
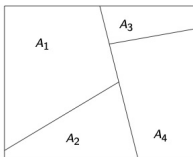
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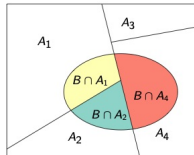
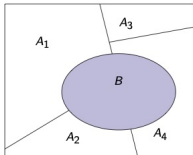
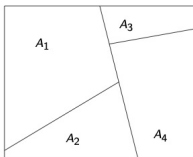
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- This is known as **total probability theorem**.

Theorem

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- Let H be the event that Allen actually has the disease. So, $P(H) = 0.0001$ and $P(H^c) = 0.999$.
- Let E be the event that the test results are positive.
- Thus, $P(E|H) = 0.99$ as test accurately find 99% of the people who have the disease.
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Probability of Allen having the disease given that the test results are positive

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Thus, these are only 9% chances that Allen has got the disease given that test results are positive.